

# Exhaustive Evidence Registry Report: E120 (Carmine, Cochineal Extract, and Carminic Acid)

## 1. Executive Conclusion

The food additive E120, universally identified as carmine, cochineal extract, or carminic acid, represents a highly stable, natural red colorant derived from the dried bodies of the female *Dactylopius coccus* insect. From a strict toxicological perspective, the global consensus among premier regulatory authorities—including the European Food Safety Authority (EFSA), the United States Food and Drug Administration (FDA), and the Joint FAO/WHO Expert Committee on Food Additives (JECFA)—is that E120 is entirely safe for general human consumption within established Acceptable Daily Intake (ADI) parameters. Extensive toxicological profiles confirm that the isolated pigment is neither genotoxic nor carcinogenic.

However, E120 introduces two critical, distinct complexities that necessitate rigorous tracking and transparent signaling within the Bari Evidence Registry. First, the additive carries a well-documented risk of immunoglobulin E (IgE)-mediated hypersensitivity and anaphylaxis. This immunological response is triggered not by the anthraquinone pigment itself, but by residual insect proteins that survive the extraction and chelation processes. While the prevalence of this allergy is statistically low in the general population, the severity of the reactions has prompted mandatory, explicit labeling requirements by the FDA, EFSA, and the Israeli Ministry of Health to ensure sensitive individuals can identify and avoid the ingredient. Second, as an unambiguously insect-derived ingredient, E120 severely conflicts with widespread consumer dietary preferences and ethical frameworks. It is strictly non-vegan and non-vegetarian, and it is broadly classified as non-kosher and haram by mainstream religious jurisprudential authorities. Because carmine is frequently marketed under the opaque umbrella of "natural colors," its presence poses a profound transparency issue. For the Bari Evidence Registry, E120 should not reduce a food's baseline nutritional score, as it is toxicologically inert, but it must trigger strict additive complexity flags, uncompromising dietary preference warnings (vegan, kosher, halal), and an objective, non-alarmist hypersensitivity caution.

## 2. Chemical Identity, Extraction, and Pharmacological Profile

### Biochemical Structure and Botanical Ecology

Carmine and cochineal extract represent distinct but biochemically related derivatives of the *Dactylopius coccus* Costa insect. This species is a sessile scale insect indigenous to the tropical and subtropical regions of Central and South America, particularly Peru and Mexico, where it feeds as a parasite on the moisture and nutrients of the *Opuntia* (prickly pear) cactus<sup>1</sup>. The primary coloring principle synthesized by the insect is carminic acid (chemical formula C<sub>22</sub>H<sub>20</sub>O<sub>13</sub>, CAS number 1260-17-9), which functions biologically as a chemical defense

mechanism against predation<sup>3</sup>.

Structurally, carminic acid is a 7-beta-D-glucopyranosyl-3,5,6,8-tetrahydroxy-1-methyl-9,10-dioxoanthracene-2-carboxylic acid<sup>4</sup>. The anthraquinone backbone of this molecule provides an exceptionally robust conjugated electron system. This structural conjugation grants carminic acid extreme resistance to thermal degradation, photodecomposition, and oxidative stress, giving it significant technological superiority over highly volatile plant-based red pigments, such as anthocyanins, betanins, and chlorophyll derivatives, which tend to degrade rapidly under varying pH environments or industrial heat treatments<sup>8</sup>.

## **Industrial Extraction and Chelation Processes**

In commercial manufacturing, the female insects are harvested, sun-dried, and pulverized. Cochineal extract is produced by subjecting the dried insect biomass to an aqueous, aqueous-alcoholic, or alcoholic extraction process. The resulting concentrated solution contains carminic acid alongside varying concentrations of insect-derived proteinaceous material and biological residues<sup>4</sup>.

Carmine, by contrast, is a more refined derivative. It is produced by subjecting carminic acid to a process of chemical transformation known as chelation, wherein the acid is precipitated with aluminum or calcium-aluminum salts to form an insoluble lake pigment<sup>1</sup>. This hydration and complexation process drastically alters the solubility profile of the dye, rendering it insoluble in cold water and dilute acids, while allowing it to be dispersed homogeneously in lipid-rich or dry matrices<sup>3</sup>. According to European Union (EU) specifications established under Commission Regulation (EU) No 231/2012, commercial preparations of pure carminic acid must contain not less than 90% of the active compound, whereas carmine lakes must contain not less than 50% carminic acid, with the remainder comprising the chelation substrate, moisture, and trace impurities<sup>5</sup>.

## **Technological Applications in the Food Supply**

Due to its robust thermodynamic stability, E120 is utilized ubiquitously across the global food, beverage, pharmaceutical, and cosmetic industries<sup>1</sup>. In the food sector, common applications include dairy desserts, strawberry and cherry-flavored yogurts, ice creams, candies, baked goods, processed meats (e.g., sausages, salami, surimi, and fish analogues), fruit juices, and alcoholic beverages (such as Campari)<sup>9</sup>. In these contexts, E120 fulfills a strictly sensory role; it is classified as a functional colorant designed to restore visual appeal lost during industrial processing or to enhance the aesthetic identity of a product. It contributes zero nutritional value, does not function as a preservative, and offers no extending effect on shelf life<sup>8</sup>.

# **3. Global Regulatory Frameworks and Toxicological Safety**

The regulatory landscape governing E120 is characterized by a universal consensus regarding its toxicological safety, juxtaposed with stringent regulatory mandates designed to manage its allergenic potential.

## **European Food Safety Authority (EFSA) Re-evaluations**

The European Food Safety Authority (EFSA) Panel on Food Additives and Nutrient Sources

Added to Food conducted an exhaustive re-evaluation of cochineal, carminic acid, and carmines in 2015. This assessment reviewed all available acute, short-term, subchronic, and long-term carcinogenicity studies conducted in animal models, including mice and rats<sup>12</sup>. The panel conclusively determined that carminic acid possesses no genotoxic, mutagenic, or carcinogenic potential<sup>5</sup>. Furthermore, comprehensive reproductive and developmental toxicity studies yielded no evidence of teratogenic effects<sup>17</sup>.

Based on the toxicological dataset, the EFSA panel concluded that there was no scientific justification to revise the previously established Acceptable Daily Intake (ADI). The ADI was reaffirmed and refined to be expressed strictly in terms of the active compound: 2.5 mg of carminic acid per kilogram of body weight per day (mg/kg bw/day), which corresponds mathematically to 5 mg of commercial carmine (containing approximately 50% carminic acid) per kg bw/day<sup>5</sup>.

EFSA also conducted refined exposure assessments across various European population demographics, utilizing food consumption data from the Comprehensive European Food Consumption Database. Under both regulatory maximum level scenarios and refined non-brand-loyal scenarios, the estimated dietary exposure to E120 remained consistently below the established ADI of 2.5 mg carminic acid/kg bw/day for all population groups, including vulnerable cohorts such as infants and toddlers<sup>17</sup>. Consequently, EFSA concluded that E120 poses no chronic or acute toxicological risk to the general public at current consumption levels.

## **United States Food and Drug Administration (FDA) Regulations**

In the United States, carmine and cochineal extract are classified as color additives exempt from the stringent batch-by-batch certification required for synthetic coal-tar dyes (such as Red 40 or Yellow 5). They are permanently listed under Title 21 of the Code of Federal Regulations (CFR) Sections 73.100 (for foods), 73.1100 (for drugs), and 73.2087 (for cosmetics)<sup>21</sup>.

Historically, U.S. regulations permitted manufacturers to obscure the presence of E120 under generic ingredient declarations such as "Artificial Color," "Color Added," or "Natural Color." However, following a sustained accumulation of adverse event reports detailing severe anaphylactic reactions, coupled with a 1998 citizen petition from the Center for Science in the Public Interest (CSPI), the FDA intervened<sup>23</sup>. In 2009, the FDA promulgated a final rule (74 FR 207) amending the CFR. The agency explicitly revoked the allowance for generic declarations and mandated that, effective January 5, 2011, all foods (including butter, cheese, and ice cream, which often enjoy labeling exemptions) and cosmetics must explicitly declare "carmine" or "cochineal extract" by name in the ingredient list<sup>23</sup>.

The FDA's rationale was heavily rooted in consumer protection for the allergic minority. While the FDA clarified that E120 does not meet the statutory definition of a "major food allergen" under the Food Allergen Labeling and Consumer Protection Act (FALCPA)—which is restricted to the top nine allergenic food groups—the agency acknowledged the undeniable clinical evidence of IgE-mediated anaphylaxis. The specific-name declaration mandate ensures that sensitized individuals can easily identify and avoid the additive<sup>23</sup>. Crucially, the FDA ruled that E120 is not exempt from labeling even if it is present merely as an incidental additive or a component of another ingredient, citing its authority under Section 403(x) of the Federal Food, Drug, and Cosmetic Act<sup>23</sup>.

## **JECFA and WHO/FAO Assessments**

The Joint FAO/WHO Expert Committee on Food Additives (JECFA) has evaluated carmine multiple times, most notably establishing an ADI of 0-5 mg/kg bw for carmines (as ammonium, calcium, potassium, or sodium salts) during its 26th meeting in 1982, an ADI that was subsequently maintained during its 55th meeting in 2000<sup>7</sup>. JECFA has established strict purity specifications for commercial E120, limiting heavy metal contaminants such as lead (maximum 1.5-2.0 mg/kg), arsenic (maximum 1 mg/kg), mercury (maximum 0.5 mg/kg), and cadmium (maximum 0.1 mg/kg)<sup>4</sup>.

Like the FDA and EFSA, JECFA explicitly noted the allergenic potential of the additive. The 55th JECFA report formally concluded that cochineal extract, carmines, and potentially carminic acid itself, when present in foods and beverages, possess the capacity to provoke severe allergic reactions in susceptible individuals<sup>6</sup>.

## **Israeli Ministry of Health Regulations**

Within the Israeli regulatory framework, E120 is permitted for use under the Public Health Regulations (Food) (Food Additives), 2001. The Israeli Ministry of Health strictly regulates the maximum permitted levels of E120 across various food categories, often clustering it alongside other synthetic dyes (such as E102, E110, and E124) with a combined maximum limit, typically capping E120 at 50 mg/kg or 50 mg/L depending on the food matrix<sup>26</sup>.

Under the Israeli standard for pre-packaged food labeling (T.I. 1145), manufacturers are legally prohibited from importing, storing, or selling any food containing a color additive unless the packaging clearly states "צבע מאכל" (Food Color) followed immediately by the specific name of the dye (e.g., כרמין) or its precise E-number (E120)<sup>27</sup>. Furthermore, Israeli regulations address the concept of "Carry Over" (תסחיב). If E120 is present in a composite food solely because it was an ingredient in a raw material, it must still comply with additive regulations, further ensuring transparent exposure tracking<sup>26</sup>.

## **4. Immunological Profiling: Hypersensitivity and Anaphylaxis**

While E120 is toxicologically benign for the general populace, it poses a distinct and highly documented immunological hazard to a specific subset of consumers. The classification of E120 as a hidden allergen is one of the primary drivers for its tracking within the Bari Evidence Registry.

### **Prevalence and Epidemiological Context**

Hypersensitivity to food additives is generally uncommon, estimated to affect between 0.01% and 0.23% of the general population. However, this prevalence rises significantly to between 2% and 7% among atopic individuals—those with a genetic predisposition to develop allergic diseases such as allergic rhinitis, asthma, or atopic dermatitis<sup>1</sup>. Within this demographic, carmine allergy represents a small but disproportionately severe clinical subset. While the overall population-level risk remains exceptionally low, over 80 severe cases of hypersensitivity—including life-threatening anaphylaxis—have been documented in the global medical literature, predominantly emerging from Japan, Europe, and the United States<sup>14</sup>.

### **The Mechanism of Action: Insect Proteins**

Clinical and immunological analyses confirm that the allergic trigger in E120 is rarely the low-molecular-weight carminic acid pigment itself. Instead, the reactions are mediated by immunoglobulin E (IgE) antibodies reacting to residual high-molecular-weight proteinaceous impurities that survive the commercial extraction process. Because commercial carmine is derived from whole insect bodies, it retains structurally distinct insect proteins. Specifically, researchers have isolated several protein bands from raw and boiled cochineal extract—ranging from 17 kDa to 130 kDa—that bind to patient serum IgE. The most prominent identified allergen is a 38 kDa protein designated CC38K, alongside other significant binding bands identified at approximately 86, 114, and 130 kDa<sup>15</sup>.

Because neither the FDA nor EFSA has established maximum legal limits on carmine protein impurities, the protein content in commercial batches can vary drastically, occasionally reaching up to 25% of the additive's mass depending on the manufacturer's purification protocols<sup>12</sup>. Consequently, EFSA has formally advised that manufacturers introduce advanced purification steps to eliminate these eliciting proteinaceous compounds, as no safe threshold dose can be established for allergic reactions<sup>17</sup>.

## **The Epicutaneous Sensitization Pathway**

A unique and highly critical aspect of carmine allergy is the epidemiological pathway of sensitization. Clinical literature reveals that many patients experiencing systemic food allergy to carmine were never sensitized through the gastrointestinal tract. Instead, initial sensitization occurs epicutaneously—through the skin—following repeated topical exposure to cosmetics containing E120<sup>14</sup>.

Carmine is ubiquitous in the cosmetic industry, heavily utilized in red and pink lipsticks, blushes, and eyeshadows due to its vibrant hue and stability. When applied to the skin—particularly damaged or highly permeable skin common in atopic individuals—the insect proteins penetrate the epidermal barrier, triggering the immune system to produce carmine-specific IgE antibodies<sup>14</sup>. A landmark retrospective analysis of carmine-induced anaphylaxis found that over 85% of documented female patients had a prior history of contact dermatitis or localized erythema (redness) induced by cosmetics before they ever experienced a food-based reaction<sup>30</sup>.

Once this epicutaneous sensitization is established, subsequent ingestion of even minute quantities of E120 in processed foods—such as a Campari soda, a red popsicle, or Japanese fish sausage (Yokan)—can provoke a massive systemic immune response, culminating in immediate anaphylaxis<sup>15</sup>. This paradigm of skin-to-gut sensitization mirrors the suspected mechanisms of other severe food allergies, such as peanut allergy, highlighting carmine as a critical hidden danger for cosmetic users<sup>30</sup>.

## **Clinical Manifestations and Diagnostic Challenges**

Reactions to E120 span a broad clinical spectrum. Occupational exposure—particularly among factory workers inhaling carmine powder during spice or dye manufacturing—has been definitively linked to the development of occupational asthma and rhinoconjunctivitis, demonstrating that the insect proteins act as potent aeroallergens<sup>29</sup>.

In the context of dietary consumption, symptoms range from chronic inducible urticaria (CIIndU), angioedema (severe swelling of the face, lips, and tongue), gastrointestinal distress,

bronchospasm, and cardiovascular collapse (anaphylaxis)<sup>14</sup>. A comprehensive 2022 study conducted at the Medical University of Gdansk investigated carmine hypersensitivity among 110 patients suffering from chronic urticaria. The study revealed that 17% of the patients exhibited a positive skin prick test (SPT) to carmine, and a placebo-controlled oral challenge confirmed definitive E120 allergy in 8% of the cohort<sup>14</sup>. The researchers noted a distinct clinical correlation: patients with carmine hypersensitivity were statistically far more likely to suffer from atopy and experience systemic symptoms (such as tongue swelling and dyspnea) compared to patients with idiopathic urticaria<sup>14</sup>.

Diagnosing carmine allergy remains exceptionally challenging because E120 is not included in standard clinical food allergy panels (which focus on milk, eggs, nuts, soy, wheat, and seafood). Patients experiencing recurrent, unexplained anaphylaxis after restaurant meals or consuming processed foods are frequently misdiagnosed with "idiopathic anaphylaxis" until a meticulous dietary history and targeted specific IgE (f340) blood tests or custom skin prick tests unveil the hidden colorant<sup>1</sup>. Furthermore, individuals with known allergies to other arthropod proteins (e.g., dust mites, shrimp, or crustaceans) may exhibit cross-reactivity due to overlapping protein structures, though studies indicate that carmine sensitization frequently occurs independently of mite allergy<sup>34</sup>.

## 5. Consumer Transparency and Dietary Preferences

Beyond its physiological impacts, E120 generates profound friction regarding consumer transparency, ethics, and religious dietary laws. The fundamental issue lies in the additive's origin: it is exclusively derived from the crushed bodies of insects. Because food manufacturers frequently market products containing E120 using opaque claims such as "No Artificial Colors" or "Contains Natural Colors," consumers rely heavily on exhaustive ingredient registries to align their diets with their ethical or religious convictions.

### Vegan and Vegetarian Incompatibility

E120 is unambiguously animal-derived. The production process requires the harvesting and systematic desiccation of tens of thousands of female *Dactylopius coccus* insects to yield even small quantities of the dye<sup>13</sup>. Consequently, any food, beverage, or cosmetic product containing E120 is strictly non-vegan and non-vegetarian. Consumers seeking plant-based alternatives must be extremely vigilant, as the regulatory designation of E120 as a "natural" colorant carries no statutory restriction against insect or animal derivation<sup>35</sup>. Legitimate vegan certifications universally exclude carmine, utilizing plant-based alternatives such as beetroot red (E162), anthocyanins (E163), or lycopene (E160d) to achieve similar hues<sup>35</sup>.

### Kashrut (Kosher) Jurisprudence

Under Jewish dietary law (Halacha), the consumption of insects and all their derivatives is strictly prohibited, categorized as *treif* (non-kosher), deriving from the biblical injunctions in Leviticus and Deuteronomy<sup>35</sup>. The status of carmine has been the subject of extensive rabbinical debate over the centuries. Some historical minority opinions, referencing arguments similar to those found in the *Noda B'Yehuda*, posited that if the insect was dried for over twelve months until it was rendered to "dust" (losing its status as an edible organism), or if the pigment was used in such minute quantities that it was nullified (*bateil*), it might be permissible<sup>36</sup>.



However, the overwhelming consensus among modern mainstream Kashrut authorities and global certification agencies (e.g., the OU, OK, Star-K, and the Israeli Chief Rabbinate) is that carmine remains strictly non-kosher<sup>35</sup>. Prominent Halachic arbiters, including Rabbi Ovadia Yosef in his responsa *Yabia Omer* and the *Minchas Yitzchok*, have affirmed the prohibition against using carmine in food<sup>36</sup>. The rationale is that since the colorant is intentionally added to impart a specific sensory characteristic (color), it holds significance and cannot be considered nullified, regardless of the dilution ratio<sup>41</sup>. Notably, this strict prohibition applies to ingestion; non-edible topical cosmetics containing carmine that have a foul taste are generally not subject to the same dietary restrictions, provided they are not swallowed<sup>45</sup>.

## Halal Jurisprudence

Islamic dietary law parallels the complexities seen in Kashrut. The permissibility of E120 is heavily debated among Islamic scholars, primarily hinging on the interpretation of insect consumption. In the Hanafi school of thought, all insects (with the strict exception of locusts) are classified as *Khabaith* (repulsive or impure) and are therefore unequivocally *Haram* (forbidden) for human consumption<sup>48</sup>. The majority opinions within the Shafi'i and Hanbali schools concur, ruling that insect-derived food additives are impermissible<sup>48</sup>.

A counter-argument utilized by some scholars and minority Halal certifiers revolves around the concept of *Istihalah* (complete chemical transformation). Proponents of this view argue that the industrial extraction, boiling, and subsequent chelation of carminic acid physically and chemically transforms the prohibited insect biomass into a distinctly new, pure substance, thereby rendering it Halal<sup>3</sup>. Furthermore, some entomological arguments suggest that because the cochineal insect is herbivorous (feeding exclusively on cacti) and bloodless, it should be exempt from standard prohibitions against scavenging insects<sup>3</sup>.

Despite these academic debates, the safest and most prevalent approach adopted by global Muslim consumers and conservative Halal certification bodies (such as JAKIM and HFA) is to reject E120 outright. Products containing carmine, cochineal, or E120 are generally avoided to maintain a strictly Halal diet<sup>37</sup>. Similar to Jewish law, the external use of carmine in topical cosmetics or clothing dyes is generally deemed permissible (*Tahir* or pure), provided there is no risk of ingestion, though products like lipsticks remain highly contested due to the probability of incidental swallowing<sup>37</sup>.

## 6. Evidence Table

The following table synthesizes the primary, high-quality scientific and regulatory sources utilized to construct the toxicological and epidemiological profile of E120.

| Source Title                  | Organization / Journal | Year | URL / DOI / PMID         | What It Says      | Evidence Strength | Registry Implication         |
|-------------------------------|------------------------|------|--------------------------|-------------------|-------------------|------------------------------|
| Scientific Opinion on the re- | European Food Safety   | 2015 | DOI: 10.2903/j.efsa.2015 | Re-evaluates E120 | High              | Confirms absolute toxicologi |

|                                                                                                                        |                                         |      |           |                                                                                                                                                                                                                |      |                                                                                                    |
|------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|----------------------------------------------------------------------------------------------------|
| evaluation of cochineal, carminic acid, carmines (E 120) as a food additive                                            | Authority (EFSA)                        |      | .4288     | safety. Establishes an ADI of 2.5 mg carminic acid/kg bw/day. Finds no genotoxicity or carcinogenicity. Notes severe risk of allergic reactions to proteinaceous impurities and advises advanced purification. |      | cal safety but requires a specific allergy caution based on protein impurities.                    |
| Listing of Color Additives Exempt From Certification; Food, Drug, and Cosmetic Labeling: Cochineal Extract and Carmine | U.S. Food and Drug Administration (FDA) | 2009 | 74 FR 207 | Final rule mandating the explicit, by-name declaration of "carmine" or "cochineal extract" on all food and cosmetic labels. Respond                                                                            | High | Mandates strict nomenclature detection for allergen tracking; prevents masking as "natural color". |



|                                                                     |                                                          |      |                     |                                                                                                                                                                        |      |                                                                                                           |
|---------------------------------------------------------------------|----------------------------------------------------------|------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----------------------------------------------------------------------------------------------------------|
|                                                                     |                                                          |      |                     | s directly to documented severe allergic reactions and anaphylaxis.                                                                                                    |      |                                                                                                           |
| Evaluation of certain food additives and contaminants (55th Report) | Joint FAO/WHO Expert Committee on Food Additives (JECFA) | 2000 | TRS 901-JECFA 55/10 | Maintains the 1982 ADI of 0-5 mg/kg bw for carmines. Formally concludes that cochineal extract and carmines may provoke allergic reactions in susceptible individuals. | High | Reinforces the international consensus on safe consumption limits paired with acknowledged allergy risks. |
| Carmine allergy in urticaria patients                               | Postepy Dermatologii i Alergologii                       | 2022 | PMID: 35369613      | Clinical epidemiological study assessing carmine hypersensitivity in 110                                                                                               | High | Provides a robust clinical baseline for the prevalence, diagnosis (IgE                                    |

|                                                                                    |                                    |      |                   |                                                                                                                                                                                                   |        |                                                                                                   |
|------------------------------------------------------------------------------------|------------------------------------|------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|---------------------------------------------------------------------------------------------------|
|                                                                                    |                                    |      |                   | patients. Finds carmine is a significant potential allergen in patients with chronic inducible urticaria (8% confirmed via oral challenge ).                                                      |        | f340), and manifestation of E120 hypersensitivity.                                                |
| Skin Sensitization to Carmine Before Onset of Systemic Allergy to Ingested Carmine | World Allergy Organization Journal | 2012 | PMCID: PMC3512794 | Investigates anaphylaxis cases due to carmine. Concludes that sensitization frequently occurs epicutaneously via cosmetics before manifesting as severe food allergy (anaphylaxis) upon ingestion | Medium | Explains the complex epidemiological mechanism of the allergy, justifying targeted caution flags. |

|                                                               |                            |                |                                   |                                                                                                                                                                              |        |                                                                                                                    |
|---------------------------------------------------------------|----------------------------|----------------|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------------------------------------------------------------------------------------------------------------------|
|                                                               |                            |                |                                   | of food.                                                                                                                                                                     |        |                                                                                                                    |
| Allergic Contact Dermatitis to Carmine                        | Dermatitis                 | 2016           | DOI: 10.1097/der.0000000000000386 | Details immediate and delayed hypersensitivity responses to carmine. Reviews patch test data and points to high-molecular-weight cochineal proteins as the allergic trigger. | Medium | Distinguishes between localized contact dermatitis and systemic IgE-mediated responses to E120.                    |
| Public Health Regulations (Food) (Food Additives) & T.I. 1145 | Israeli Ministry of Health | 2001 (updated) | MoH Registry / T.I. 1145          | Lists E120 as a permitted food colorant with maximum permitted levels depending on the food category. Mandates explicit labeling under                                       | High   | Dictates local legal compliance and validates the Hebrew nomenclature detection patterns utilized in the registry. |

|                                                               |                      |      |                                 |                                                                                                                                                                               |        |                                                                                                                                      |
|---------------------------------------------------------------|----------------------|------|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------|
|                                                               |                      |      |                                 | Israeli standard T.I. 1145, including "Carry Over" rules.                                                                                                                     |        |                                                                                                                                      |
| Unexplained Anaphylaxis Unveiled: The Role of Carmine Allergy | Frontiers in Allergy | 2024 | DOI: 10.3389/falgy.2025.1661992 | Case report of recurrent idiopathic anaphylaxis eventually traced to carmine food additive in restaurant food. Emphasizes the need for distinct labeling of hidden allergens. | Medium | Highlights the severe clinical difficulty in identifying E120 allergy and the profound necessity of transparent consumer registries. |

## 7. Bari Evidence Registry Proposal

The following structured data arrays are formulated for direct, programmatic integration into the Bari Evidence Registry database.

| Field     | Value                 |
|-----------|-----------------------|
| signal_id | additive_e120_carmine |

|                                |                                                                                                                                                                                         |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ingredient_names               | E120, Carmine, Cochineal Extract, Carminic Acid, Natural Red 4, CI 75470                                                                                                                |
| Hebrew patterns                | `(?i)(E-?120                                                                                                                                                                            |
| English patterns               | `(?i)(E-?120                                                                                                                                                                            |
| additive_class                 | colorant                                                                                                                                                                                |
| evidence_level                 | high                                                                                                                                                                                    |
| suggested_user_facing_label_he | מופק מחרקים – (E120) צבע מאכל אדום                                                                                                                                                      |
| suggested_explanation_he       | צבע מאכל טבעי המופק מחרקים (כנימת קוצ'יניל). בטוח לצריכה לרוב האוכלוסייה, אך אינו מתאים לטבעונים, נחשב לרוב כלא-כשר או חראם, ועלול לגרום לתגובות אלרגיות חמורות בקרב אנשים בעלי רגישות. |
| suggested_scoring_effect       | neutral_nutrition_high_transparency_flag                                                                                                                                                |
| suggested_confidence           | 1.0                                                                                                                                                                                     |
| citations                      | [cite: 15, 17, 18, 23, 35]                                                                                                                                                              |

## 8. Detection Patterns and Nomenclature Morphology

To ensure the Bari Evidence Registry captures E120 across highly variable global and localized supply chains, robust detection algorithms must be implemented. Manufacturers employ a wide spectrum of nomenclatures—ranging from chemical identifiers and regulatory codes to colloquial names—often in an attempt to obscure the ingredient's origin.

- **E-Number and Regulatory Variations:** Regular expressions (regex) must account for standard European and international identifiers, capturing exact matches for E120, E-120, E 120, and the Codex Alimentarius equivalent INS 120.
- **English Nomenclatures:** The detection logic must encompass carmine, carmines, carminic acid, cochineal, cochineal extract, natural red 4, and the cosmetic/color index

identifier CI 75470. It is critical to note that the generic phrase "natural red color" alone is insufficient to guarantee the presence of E120, as it may refer to plant-based alternatives like beetroot extract or anthocyanins. However, exact matches to the aforementioned specific terms are definitive.

- **Hebrew Nomenclatures:** Israeli food labeling standards (T.I. 1145) require the explicit declaration of E-numbers or specific Hebrew names following the prefix "צבע מאכל" (Food Color). The algorithmic match patterns must include:
  - כרמין (Carmine)
  - קרמין (Carmine - an alternative phonetic spelling)
  - קוצ'יניל / קוצ'יניל (Cochineal, accounting for typographic variations in the apostrophe/geresh)
  - חומצה כרמינית (Carminic acid)
  - אדום טבעי 4 (Natural red 4)

Algorithmically, exact substring matching bounded by word borders or comma separation in digitized ingredient lists is highly recommended to prevent false positives and ensure accurate flagging.

## 9. Scoring Recommendations and Algorithm Logic

To accurately reflect the complex duality of E120—toxicologically safe yet ethically and immunologically fraught—the Bari system should process the additive using a multi-tiered algorithmic approach.

### Nutritional Baseline Impact

**Should E120 reduce the Bari nutrition score? No.** Extensive, multi-generational toxicological reviews conducted by EFSA, the FDA, and JECFA universally confirm that E120 possesses no negative metabolic, genotoxic, carcinogenic, or chronic toxicity effects at established exposure levels<sup>5</sup>. Penalizing a food product's baseline nutritional quality score solely due to the presence of E120 would be scientifically inaccurate and would misalign the Bari registry with global toxicological consensus. The additive is metabolically inert.

### Additive and Ultra-Processed Food (UPF) Complexity

**Should it increase additive/UPF complexity? Yes.** E120 is a highly refined, industrially extracted, and chemically chelated cosmetic additive. It is utilized exclusively to alter the sensory properties of food without imparting any nutritional value. Consequently, its presence is a definitive marker of ultra-processed foods (aligning with NOVA Group 4 classifications). It indicates a high degree of industrial formulation and cosmetic manipulation, warranting an increase in the product's processing complexity score.

### Dietary-Preference Transparency

**Should it trigger a dietary-preference transparency flag? Yes, this requires a Critical Flag.** This is arguably the most vital consumer-facing function for E120 within the registry. The system must execute hard-flags designating E120 as **Non-Vegan**, **Non-Vegetarian**, **Non-Kosher**, and **Non-Halal** (subject to the user's specific preference settings). Because E120 is frequently disguised under the benign, highly appealing term "natural color," the application provides immense value by piercing this opacity. Revealing the insect origin of the dye empowers



consumers guided by strict ethical, philosophical, or religious frameworks to make informed purchasing decisions<sup>35</sup>.

## Immunological Caution

**Should it trigger an allergy/hypersensitivity caution? Yes, this requires a targeted**

**Caution Flag.** E120 must trigger an explicit allergen warning. Because E120 is an atypical allergen—meaning it is not covered by the standard "top-9" or "top-14" major allergen bolding requirements mandated by the FDA or EU—allergic users are entirely reliant on meticulous label reading and third-party registries to prevent anaphylaxis<sup>23</sup>. The registry should explicitly note that the allergy is triggered by residual insect proteins. To prevent alert fatigue among the general user base, this specific flag should ideally be reserved for users who have activated hypersensitivity or allergy settings in their profiles.

## Proposed User-Facing Wording

The wording presented to the user must remain strictly objective, scientifically accurate, and descriptive, actively avoiding toxicological alarmism while maximizing practical consumer transparency.

- **English:** "E120 (Carmine / Cochineal Extract) is a natural red colorant derived from the crushed bodies of insects. While global health authorities recognize it as toxicologically safe for general consumption, it is strictly non-vegan and is widely classified as non-kosher and haram. Additionally, residual insect proteins in the dye can cause severe allergic reactions, including anaphylaxis, in sensitive individuals."
- **Hebrew:** "המופק מגופן של כנימות (חרקים). על אף שהוא מוגדר כבטוח (E120) צבע מאכל טבעי, לחלוטין לצריכה על ידי רשויות הבריאות בעולם, הוא אינו טבעוני, נחשב לרוב כלא-כשר וכחראם, ועלול לעורר תגובות אלרגיות חמורות בקרב אנשים בעלי רגישות לחלבוני חרקים."

## Works cited

1. Case Report: Not all recurrent "idiopathic" anaphylaxis is idiopathic - PMC, <https://pmc.ncbi.nlm.nih.gov/articles/PMC12528176/>
2. Case Report: Not all recurrent "idiopathic" anaphylaxis is idiopathic - Frontiers, <https://www.frontiersin.org/journals/allergy/articles/10.3389/falgy.2025.1661992/full>
3. Reevaluating the halal status of Carmine (E120): A scientific and Islamic legal review, [https://www.researchgate.net/publication/394167336\\_Reevaluating\\_the\\_halal\\_status\\_of\\_Carmine\\_E120\\_A\\_scientific\\_and\\_Islamic\\_legal\\_review](https://www.researchgate.net/publication/394167336_Reevaluating_the_halal_status_of_Carmine_E120_A_scientific_and_Islamic_legal_review)
4. COCHINEAL EXTRACT, [https://www.fao.org/fileadmin/user\\_upload/jecfa\\_additives/docs/Monograph1/Additive-137.pdf](https://www.fao.org/fileadmin/user_upload/jecfa_additives/docs/Monograph1/Additive-137.pdf)
5. Safety and efficacy of a feed additive consisting of carmine for cats and dogs (Mars Petcare GmbH) - PMC, <https://pmc.ncbi.nlm.nih.gov/articles/PMC9598894/>
6. cochineal and carminic acid - WHO | JECFA, <https://apps.who.int/food-additives-contaminants-jecfa-database/Home/Chemical/1507>
7. cochineal carmine - WHO | JECFA, <https://apps.who.int/food-additives-contaminants-jecfa-database/Home/Chemical/1508>

8. Insect-derived food colorants: Chemistry, applications, safety, and regulatory perspectives - Agronomy Journal, <https://www.agronomyjournals.com/archives/2026/vol9issue2/PartF/9-2-79-742.pdf>
9. Food Colour Additives: A Synoptical Overview on Their Chemical Properties, Applications in Food Products, and Health Side Effects - MDPI, <https://www.mdpi.com/2304-8158/11/3/379>
10. Carmines - International Association of Color Manufacturers, <https://iacmcolor.org/color-profile/carmines/>
11. Safety and efficacy of a feed additive consisting of carmine for cats and dogs (Mars Petcare GmbH) - - 2022 - EFSA Journal, <https://efsa.onlinelibrary.wiley.com/doi/10.2903/j.efsa.2022.7609>
12. Commission Regulation - Legislation.gov.uk, <https://www.legislation.gov.uk/eur/2018/1472/data.html>
13. Cochineal Extract - KidsAdvisory, <https://kidsadvisory.com/ingredients/cochineal-extract/>
14. Carmine allergy in urticaria patients - PMC - NIH, <https://pmc.ncbi.nlm.nih.gov/articles/PMC8953891/>
15. A Case of Fish Sausage Anaphylaxis Induced by Epicutaneous Sensitization to Carmine Contained in Eyeshadows - PMC, <https://pmc.ncbi.nlm.nih.gov/articles/PMC11410408/>
16. Guidance on Food Additives, <https://www.fsai.ie/getmedia/7a4f1825-3423-45cf-ba32-0154847fdb8/food-additives-report-2015-final.pdf?ext=.pdf>
17. Scientific Opinion on the re-evaluation of cochineal, carminic acid, carmines (E 120) as a food additive - PubMed, <https://pubmed.ncbi.nlm.nih.gov/42169840/>
18. Scientific Opinion on the re-evaluation of cochineal, carminic acid, carmines (E 120) as a food additive - - 2015 - EFSA Journal, <https://efsa.onlinelibrary.wiley.com/doi/abs/10.2903/j.efsa.2015.4288>
19. Carmine allergy in urticaria patients - Termedia, <https://www.termedia.pl/Journal/-7/pdf-42400-10>
20. Dietary exposure to lycopene from background diet, novel foods and food additive use in the European population - EFSA Journal, <https://efsa.onlinelibrary.wiley.com/doi/pdf/10.2903/j.efsa.2026.9994>
21. Regulatory Status of Color Additives - FDA, <https://www.fda.gov/ForIndustry/ColorAdditives/ColorAdditiveInventories/ucm106626.htm>
22. Cochineal Extract (Carminic Acid) - hfpappexternal.fda., <https://www.hfpappexternal.fda.gov/scripts/fdcc/index.cfm?set=ColorAdditives&id=CochinealExtract>
23. Listing of Color Additives Exempt From Certification; Food, Drug, and Cosmetic Labeling: Cochineal Extract and Carmine Declaration - Federal Register, <https://www.federalregister.gov/documents/2009/01/05/E8-31253/listing-of-color-additives-exempt-from-certification-food-drug-and-cosmetic-labeling-cochineal>
24. Listing of Color Additives Exempt From Certification; Food, Drug, and Cosmetic Labeling: Cochineal Extract and Carmine Declaration; Confirmation of Effective Date - Regulations.gov, <https://www.regulations.gov/document/FDA-1998-P-0032->

0006

25. Small Entity Compliance Guide: Declaration by Name on the Label of All Foods and Cosmetic Products that Contain Cochineal Extract and Carmine | FDA, <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/small-entity-compliance-guide-declaration-name-label-all-foods-and-cosmetic-products-contain>
26. Gov, רשימת תוספי מזון פי תקנות בריאות הציבור, [https://www.gov.il/BlobFolder/reports/fcs\\_list/he/files\\_publications\\_food\\_food-addictive-library\\_fcs\\_list2017.pdf](https://www.gov.il/BlobFolder/reports/fcs_list/he/files_publications_food_food-addictive-library_fcs_list2017.pdf)
27. ויקיטקסט - (מזון) (צבעים במזון) - תקנות בריאות הציבור, [https://he.wikisource.org/wiki/%D7%AA%D7%A7%D7%A0%D7%95%D7%AA\\_%D7%91%D7%A8%D7%99%D7%90%D7%95%D7%AA\\_%D7%94%D7%A6%D7%99%D7%91%D7%95%D7%A8\\_\(%D7%9E%D7%96%D7%95%D7%9F\)\\_\(%D7%A6%D7%91%D7%A2%D7%99%D7%9D\\_%D7%91%D7%9E%D7%96%D7%95%D7%9F\)](https://he.wikisource.org/wiki/%D7%AA%D7%A7%D7%A0%D7%95%D7%AA_%D7%91%D7%A8%D7%99%D7%90%D7%95%D7%AA_%D7%94%D7%A6%D7%99%D7%91%D7%95%D7%A8_(%D7%9E%D7%96%D7%95%D7%9F)_(%D7%A6%D7%91%D7%A2%D7%99%D7%9D_%D7%91%D7%9E%D7%96%D7%95%D7%9F))
28. תזכיר תקנות הגנה על בריאות הציבור (מזון)(תוספי מזון), התשפ"א – 2021 - תקדין, <https://www.takdin.co.il/document/index/7011251-%D7%AA%D7%96%D7%9B%D7%99%D7%A8-%D7%AA%D7%A7%D7%A0%D7%95%D7%AA-%D7%94%D7%92%D7%A0%D7%94-%D7%A2%D7%9C-%D7%91%D7%A8%D7%99%D7%90%D7%95%D7%AA-%D7%94%D7%A6%D7%99%D7%91%D7%95%D7%A8-%D7%9E%D7%96%D7%95%D7%9F%D7%AA%D7%95%D7%A1%D7%A4%D7%99-%D7%9E%D7%96%D7%95%D7%9F-%D7%94%D7%AA%D7%A9%D7%A4%D7%90-2021>
29. f340 Cochineal extract (Carmine red) - Thermo Fisher Scientific, <https://www.thermofisher.com/phadia/wo/en/resources/allergen-encyclopedia/f340.html>
30. 494 Skin Sensitization to Carmine Before Onset of Systemic Allergy to Ingested Carmine, <https://pmc.ncbi.nlm.nih.gov/articles/PMC3512794/>
31. [Asthma and allergy due to carmine dye] - PubMed, <https://pubmed.ncbi.nlm.nih.gov/13679965/>
32. Carmine allergy in urticaria patients - PubMed, <https://pubmed.ncbi.nlm.nih.gov/35369613/>
33. Case Report: Not all recurrent "idiopathic" anaphylaxis is idiopathic - PubMed, <https://pubmed.ncbi.nlm.nih.gov/41113612/>
34. Allergy to carmine red (E120) is not dependent on concurrent mite allergy - PubMed, <https://pubmed.ncbi.nlm.nih.gov/19439984/>
35. Carmine: The Insect-Derived Red Dye Hiding in Your Food - IngrediCheck, <https://www.ingredicheck.app/blog/carmine-the-insect-derived-red-dye-hiding-in-your-food>
36. Food Colorants - Kosher Spirit, <https://indiakoshercertification.in/kosherspirit/winter-2014/food-colorants/>
37. Carmine in Makeup & Skincare: Why It's Not Halal and What to Use Instead - Purelab Care, <https://purelabcare.com/blogs/ingredients-uncovered-and-is-it-halal/carmine-in-makeup-skincare-why-it-s-not-halal-and-what-to-use-instead>
38. Is E120 (Cochineal / Carmines) Halal or Haram? - E-Code Halal Check, <https://www.ecodehalalcheck.com/ecode/120>

39. Can Muslims Eat Carmine? Who guarantee Halal certification? - Imbarex, <https://imbarex.com/can-muslims-eat-carmine/>
40. Kosher foods - Wikipedia, [https://en.wikipedia.org/wiki/Kosher\\_foods](https://en.wikipedia.org/wiki/Kosher_foods)
41. Observing a Colorful Lifestyle | Rabbi Yirmiyohu Kaganoff | Beit Midrash | yeshiva.co, <https://www.yeshiva.co/midrash/15537>
42. Most Reputable Kosher, <https://www.holykosher.com/most-reputable-kosher.html>
43. מה מסתתר ברשימת חומרי הגלם? | כושרות - ייעוץ והכוונה בכשרות, <https://www.kosharot.co.il/gelem>
44. כשרות Archives - אמס, <https://www.emess.co.il/radio/tag/%D7%9B%D7%A9%D7%A8%D7%95%D7%AA/feed>
45. Halachic and Hashkafic Issues OU - Series 2 - Shiur 18 - Cosmetics on Pesach.dwd, <https://www.ou.org/holidays/files/Halachic-and-Hashkafic-Issues-OU-Series-2-Shiur-18-Cosmetics-on-Pesach.pdf>
46. האם צריך כשרות על מוצרי איפור? - הידברות, <https://www.hidabroot.org/question/255882>
47. האם מותר לקרוא ספר אחרי קריאת שמע? האם מותר ללמוד בלי גרביים? והאם מותר לקנות בגדים בשלושת השבועות? - הידברות, <https://www.hidabroot.org/article/1127393>
48. Is E120 (Carmine) Halal? The Crushed Bug Trap Explained (2026) - Tayib, <https://tayib.app/blog/is-e120-halal-shocking-truth-about-carmine/>
49. Is Carmine Halal? (Hanafi) | Islam Q&A - Jibreel App, <https://www.jibreel.app/islamqa/is-carmine-halal>
50. Investigation of Halal Status and Permitted Industrial Utilization of Carmine Dye in Iran, [https://www.researchgate.net/publication/389114193\\_Investigation\\_of\\_Halal\\_Status\\_and\\_Permitted\\_Industrial\\_Utilization\\_of\\_Carmine\\_Dye\\_in\\_Iran](https://www.researchgate.net/publication/389114193_Investigation_of_Halal_Status_and_Permitted_Industrial_Utilization_of_Carmine_Dye_in_Iran)